Fiscal Year:	FY 2016	**Task Last Updated:**	FY 05/18/2017
PI Name:	Voorhees, Peter W. Ph.D.		
Project Title:	Coarsening of Dendritic Solid-Liquid Mixtures: The Low Volume Fraction Limit (NNX14AB73G)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	60208-0893	**Congressional District:**	9
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2010 Materials Science NNH10ZTT001N
Start Date:	01/31/2014	**End Date:**	07/31/2016
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
Contact Monitor:	Rogers, Jan	**Contact Phone:**	256.544.1081
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Extended to 7/31/2016 per NSSC information (Ed., 3/14/16)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX14AB73G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE (Ed., February 2014): Continuation of "Coarsening of Dendritic Solid-Liquid Mixtures: The Low Volume Fraction Limit", grant # NNX10AV36G with the same Principal Investigator. Dendrites form during solidification into an undercooled melt. These dendrites possess secondary and sometimes even ternary arms. While the tip radius and tip velocity of the dendrite are set by the growth conditions, the sidebranches behind the tip undergo a coarsening process under nearly isothermal conditions. This coarsening process sets the final arm thicknesses and distances between dendrite arms in the solidified structure - almost independent of the length-scale given by the dendrite tip. Unfortunately, the morphological evolution of these secondary dendrite arms remains poorly understood. Since there is a close relationship between the size scale of the coarsened dendritic structure and the mechanical properties of the material, a greater understanding of this process will have important practical implications. The reduced gravitational accelerations found during processing on the International Space Station can allow for unprecedented insights into coarsening processes of solid-liquid mixtures in which the volume fraction of solid dendrites is small. In a microgravity environment it will be possible to follow the evolution of the morphology of the mixtures in the diffusive limit where the concentration field is set by the curvatures of the solid-liquid interfaces. It will also be possible to determine the evolution of the topology of the solid-liquid mixtures. For example, when secondary dendrite arms detach from the primary dendrite stem they will not sediment, and it will thus be possible to determine the rate at which this detachment process occurs and its spatial location. The topology and morphology of these dendritic mixtures will be determined using three-dimensional reconstructions and insights into the mechanisms of coarsening will be provided by phase field simulations that employ these reconstructions as initial conditions. These studies will provide insights into the dynamics of morphological evolution of these systems and will point the way to the formulation of models of this technologically important coarsening process.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Dendritic solid-liquid mixtures are ideal topologically complex systems with which to study coarsening. The microstructure is on a size scale that is amenable to experimental analysis using three-dimensional reconstructions and the thermophysical properties of many alloy systems are reasonably well known. Moreover, the coarsening of dendritic solid-liquid mixtures has major technological implications since it occurs in virtually all casting processes. In this case, a dendritic structure forms quickly and a so-called "mushy zone" is created. This mushy zone can exist often for long periods of time during which virtually every dendrite undergoes coarsening. The length-scales of the dendrites in the mushy zone, and thus those present after complete solidification, are determined by the coarsening process. These length scales set the microsegregation pattern and are intimately related to the mechanical properties of the ingot. Understanding the dynamics of coarsening will enable a close connection to be made between solidification conditions and the properties of the material.
Task Progress:	Project continues with "Coarsening of Dendritic Solid-Liquid Mixtures (CSLM): The Low Volume Fraction Limit (NNX16AR13G)", grant NNX16AR13G with the same Principal Investigator, Dr. Peter Voorhees. See that project for subsequent reporting.
Bibliography Type:	Description: (Last Updated: 01/17/2024)